

Project Design Phase
Problem – Solution Fit

Date	29-06-2026
Team ID	23A51A0212
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Solution Fit:

The Metro Ticket Booking System developed on the ServiceNow platform provides a digital and automated solution to overcome the challenges of traditional metro ticketing. The system enables passengers to book tickets online by selecting source and destination stations, journey date, and passenger count through a user-friendly Service Catalog interface.

The solution automatically calculates fares based on predefined route information, eliminating manual calculations and reducing errors. Upon submission, the system generates a digital ticket, creates a request record, and sends booking confirmation notifications to the user. The integration of QR-code-based ticket generation further enhances convenience by enabling contactless ticket verification.

Additionally, administrators can manage metro stations, fare details, and monitor ticket booking trends through dashboards and reports. By digitizing the entire ticketing workflow, the solution reduces waiting time, improves operational efficiency, enhances user experience, and supports scalable metro transportation services.

Define CS, fit into CC	1. CUSTOMER SEGMENT(CS) CS - Daily metro commuters [cite: 1] - Office employees [cite: 1] - College students [cite: 1] - Occasional travelers [cite: 1] - Metro administration staff [cite: 1]	6. CUSTOMER CONSTRAINTS CC - Peak-hour congestion [cite: 6] - Long waiting times [cite: 6] - Lack of digital ticketing options [cite: 6] - Manual fare calculation [cite: 6] - Risk of losing physical tickets [cite: 6] - Limited access to ticket counters [cite: 6]	5. AVAILABLE SOLUTIONS AS Current alternatives used by customers: - Physical ticket counters [cite: 5] - Ticket vending machines [cite: 5] - Smart cards [cite: 5] - Manual fare enquiry [cite: 5] - Mobile payment at counters [cite: 5] Limitations: - Long queues [cite: 5] - Time-consuming process [cite: 5] - Dependence on physical infrastructure [cite: 5] - Limited accessibility [cite: 5]	Explore AS, differentiate
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Book metro tickets quickly [cite: 2] - Avoid waiting in long queues [cite: 2] - Know the exact fare before travel [cite: 2] - Access tickets digitally [cite: 2] - Complete ticket booking from anywhere 3. TRIGGERS - Need to travel during peak hours [cite: 3] - Long queues at ticket counters [cite: 3] - Availability of digital booking services [cite: 3] - Desire for faster and contactless ticketing [cite: 3] - Convenience of online booking [cite: 3]	9. PROBLEM ROOT CAUSE RC - Dependence on manual ticket booking [cite: 9] - Lack of centralized digital ticketing [cite: 9] - Manual fare calculation processes [cite: 9] - High passenger volume during peak hours [cite: 9] - Limited automation in ticket management [cite: 9]	7. BEHAVIOUR BE Alternatives/Limitations - Prefer quick and convenient services [cite: 7] - Use smartphones for daily activities [cite: 7] - Adopt digital payment methods [cite: 7] - Expect real-time information [cite: 7] - Choose self-service solutions whenever available [cite: 7]	Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	4. Before: TR - Frustrated by long waiting times [cite: 4] - Confused about fare calculations [cite: 4] - Worried about missing trains [cite: 4] - Inconvenienced by physical ticketing [cite: 4] 4. After: EM - Satisfied with quick booking [cite: 4] - Confident about fare accuracy [cite: 4] - Comfortable using digital tickets [cite: 4] - Improved travel experience [cite: 4]	10. YOUR SOLUTION SL - Online metro ticket booking through Service Catalog [cite: 10] - Automatic fare calculation [cite: 10] - Instant ticket generation (REQ/RITM) [cite: 10] - QR-code-based digital ticketing [cite: 10] - Automated email notifications [cite: 10] - Real-time dashboards and booking analytics [cite: 10] - Reduced waiting time [cite: 10]	8. CHANNELS OF BEHAVIOUR CH Online - ServiceNow Service Portal [cite: 8] - Web browsers [cite: 8] - Mobile devices [cite: 8] - Email notifications [cite: 8] Offline - Metro stations [cite: 8] - Ticket counters [cite: 8] - Metro information kiosks [cite: 8] - Customer support desks [cite: 8]	Extract online & offline CH of BE